

SDA Matrimony Platform — Development Session Summary

Date: September 20, 2026 | Branch: [email-&-authentication](#) | Commits: [aeb87a6](#), [5554427](#)

Executive Summary

During this engineering session, we implemented and deployed two major backend modules and drafted an architectural blueprint for the pastor administration portal on the NestJS core backend (apps/api-core):

- **Email Verification & Forgot Password:** Full security lifecycle with cryptographically secure SHA-256 token hashing, 1-hour verification expiry, 30-minute password reset expiry, and an automated transactional mailer.
- **Pastor Admin Panel Audit & Roadmap:** Architectural evaluation of the 7 core pastoral duties, filtering into a focused 5-pillar design (excluding counseling and community complaints as specified).
- **Admin-Assisted Matchmaking Module:** End-to-end admin suggestion engine, custom Prisma relations, strict authorization response endpoints, and standalone migration SQL.

1. Email Verification & Password Recovery Module

Database & Schema Changes

Added emailVerifyExpires DateTime? directly beneath emailVerifyToken in packages/database/prisma/schema.prisma on the User model. Prepared standalone SQL migration:
packages/database/prisma/migrations/20260919180000_add_email_verify_expiry/migration.sql.

Transactional Mailer Service (apps/api-core/src/modules/mail)

Installed nodemailer and @types/nodemailer. Configured MailService reading SMTP settings (SMTP_HOST, SMTP_PORT, SMTP_USER, SMTP_PASSWORD, SMTP_FROM_EMAIL, SMTP_FROM_NAME). Includes development fallback logging verification/reset links directly to console if SMTP is unconfigured.

Security & Cryptographic Standards

- **Token Generation:** 32-byte secure random hex string generated via crypto.randomBytes(32).toString('hex').
- **Database Storage:** Raw tokens are never stored; only SHA-256 hashes (crypto.createHash('sha256').digest('hex')) are stored, preventing disclosure during potential database compromises while maintaining fast O(1) indexed lookup.
- **Anti-Enumeration:** Forgot-password endpoint always returns a generic success message regardless of email existence to prevent user enumeration attacks.

Authentication Endpoints Added

Method	Endpoint	Access	Summary
POST	/api/v1/auth/register	Public	Registers user, hashes verify token (1h expiry), sends email
POST	/api/v1/auth/verify-email	Public	Validates token and expiry, marks isEmailVerified: true
POST	/api/v1/auth/forgot-password	Public	Generates 30m reset token, emails user (anti-enumeration)
POST	/api/v1/auth/reset-password	Public	Validates token & expiry, hashes new password with bcrypt

2. Pastor Admin Panel Blueprint & Scope

Following detailed requirements review, spiritual counseling (#3) and community complaints (#5) were permanently excluded. The retained 5 pillars focus on ecclesiastical verification and marriage coordination:

- 1. **Member Verification:** Direct token endorsement link (/pastor/endorse/[token]) for frictionless verification without account login + PDF Pastor's Certificate generation.
- 2. **Profile Review:** Local church review queue for PASTOR_VERIFIER to verify candidate details prior to public matchmaking.
- 3. **Trusted Mediation:** Pastoral introduction assistance and marital status clearance for sensitive cases (Divorced/Widowed).
- 4. **Church Coordination:** Generation of official church marriage documents (No Objection Certificates, Banns of Marriage announcements, wedding ceremony coordination).
- 5. **Privacy & Discretion:** Confidential pastoral notes invisible to candidates and suitors.

3. Admin-Assisted Matchmaking Module

Created a complete admin-assisted matchmaking system in apps/api-core/src/modules/matchmaking with custom Prisma models and DTO validations:

- **Prisma Model:** MatchSuggestion with adminId, userId, suggestedUserId, status (enum: PENDING, ACCEPTED, REJECTED), adminNote, and timestamps.
- **User Relations:** createdMatchSuggestions, receivedMatchSuggestions, suggestedAsMatch with distinct named relations.
- **Authorization & Ownership:** Recipient member must match userId to respond (throws ForbiddenException otherwise). Prevents self-suggestions and duplicate pending suggestions.

Matchmaking Endpoints Added

Method	Endpoint	Access	Summary
POST	/api/v1/admin/matches/suggest	Admin Only	Creates match suggestion with PENDING status and admin note
GET	/api/v1/admin/matches	Admin Only	Lists all suggestions with user & candidate profile summaries
GET	/api/v1/matches/suggested	Auth Member	Returns all pending suggestions for the logged-in candidate
POST	/api/v1/matches/:id/respond	Auth Member	Accepts or rejects suggestion (enforces ownership guard)

4. Database Migrations & Version Control

Standalone SQL migration scripts prepared for deployment once live PostgreSQL is connected:

- 1. packages/database/prisma/migrations/20260919180000_add_email_verify_expiry/migration.sql
- 2. packages/database/prisma/migrations/20260920003000_add_match_suggestion/migration.sql

Git Branch: email-&-authentication | **Remote:** Up to date with origin

- **Commit aeb87a6:** feat(auth): add email verification and forgot-password features
- **Commit 5554427:** feat(matchmaking): add admin-assisted matchmaking module and MatchSuggestion schema